In-Lab Tasks: Function Practice

COURSE CODE: C1002

SUBJECT: PROGRAMMING FUNDAMENTALS

INSTRUCTOR: MR. SHAYAN ALI

NAME: AMNA KHAN

ROLL NO/ID: 26K-3091

CLASS/SECTION: C

DATE OF SUBMISSION: 22-OCT'2026

Task 1 (ATM Management System Using Functions):

Write a C program that simulates a simple ATM system using five user-defined functions. Create welcome(), which takes no arguments and returns nothing,to display a welcome message. Create displayAmount(), which accepts a withdrawal amount, returns nothing, and displays the amount. Create getWithdrawalLimit(), which takes no arguments and returns the predefined maximum withdrawal limit of Rs. 50,000. Create calculateBalance(), which accepts the current account balance and the withdrawal amount and returns the remaining balance. Create checkWithdrawal(), which accepts the current balance, withdrawal amount, and maximum limit, and returns 1 if the withdrawal is valid or 0 otherwise. A withdrawal is valid only if the amount is greater than zero,does not exceed the account balance, and does not exceed the maximum withdrawal limit. In main(), call the functions as required, display the relevant information, and update the balance only if the withdrawal is valid.

Untitled1.cpp

```c
#include <stdio.h>

void welcome();
void displayAmount(float amount);
float getWithdrawalLimit();
float calculateBalance(float currentBalance, float withdrawalAmount);
int checkWithdrawal(float currentBalance, float withdrawalAmount, float maxLimit);

int main() {
    float currentBalance = 65000;
    float withdrawAmount;
    float maxLimit;

    maxLimit = getWithdrawalLimit();

    welcome();

    printf("Your balance is: Rs. %.1f\n", currentBalance);
    printf("Enter amount to withdraw: ");
    scanf("%f", &withdrawAmount);

    if (checkWithdrawal(currentBalance, withdrawAmount, maxLimit) == 1) {
        displayAmount(withdrawAmount);
        currentBalance = calculateBalance(currentBalance, withdrawAmount);
        printf("Remaining balance: Rs. %.1f\n", currentBalance);
    } else {
        printf("Withdrawal failed.\n");
    }

    return 0;
}

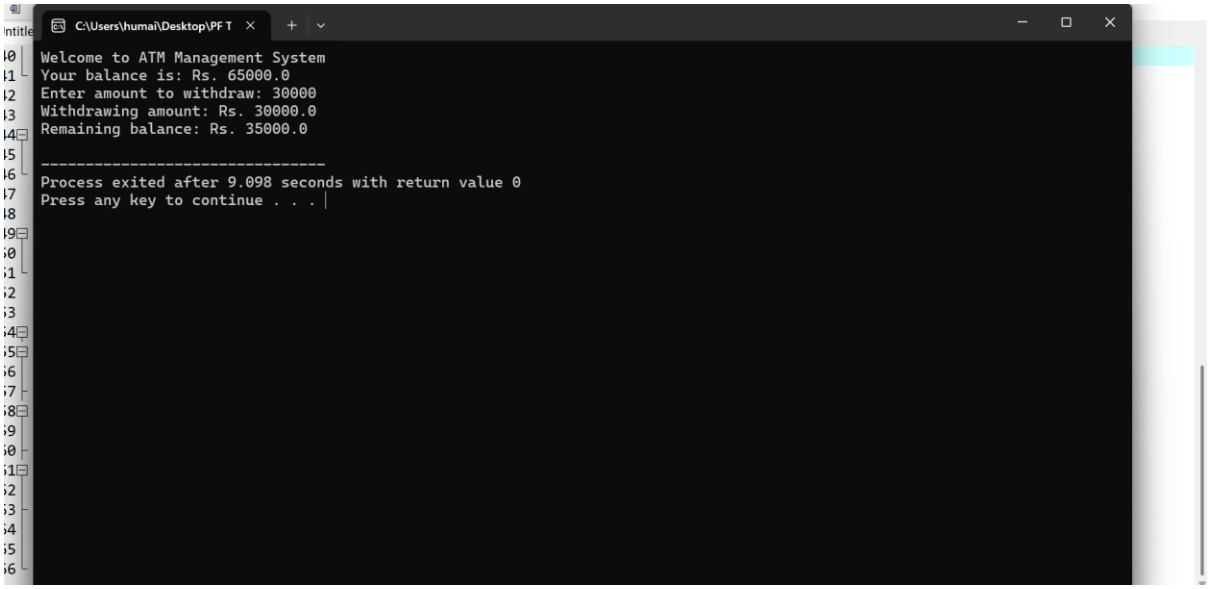
// Function 1
void welcome() {
    printf("Welcome to ATM Management System\n");
}

// Function 2
void displayAmount(float amount) {
    printf("Withdrawing amount: Rs. %.1f\n", amount);
}
// Function 3
float getWithdrawalLimit() {
    return 50000;
}

// Function 4
float calculateBalance(float currentBalance, float withdrawalAmount) {
    return currentBalance - withdrawalAmount;
}

// Function 5
int checkWithdrawal(float currentBalance, float withdrawalAmount, float maxLimit) {
    if (withdrawalAmount <= 0) {
        return 0;
    }
    if (withdrawalAmount > maxLimit) {
        return 0;
    }
    if (withdrawalAmount > currentBalance) {
        return 0;
    }

    return 1;
}
```



Task 2 (Shopping Management System Using Functions):

Write a C program that simulates a shopping management system using user-defined functions. The program should display a welcome message, show a menu of available products with their prices, allow the user to select products and enter quantities, calculate the total price of each product, calculate the subtotal of all selected products, apply a discount if the subtotal reaches a specified amount, calculate the final bill, and display a shopping receipt. Create separate functions to display the welcome message, show the product menu, get the product price, calculate the price of an item, calculate the subtotal, calculate the discount, calculate the final bill, display the receipt, and ask the user whether they want to continue shopping. Use functions with different combinations of arguments and return types, including functions that take no arguments and return no value, functions that take arguments and return no value, functions that take no arguments and return a value, and functions that take arguments and return a value. Use appropriate loops and conditional statements to manage product selection, quantities, discounts, and repeated shopping. Display the selected products, quantities, individual prices, subtotal,discount, and final payable amount.

Untitled1.cpp

```c
#include <stdio.h>

// 1.
void showMenu() {
    printf("\n=== SHOPPING MENU ===\n");
    printf("1. Laptop      - Rs. 50,000\n");
    printf("2. Smartphone  - Rs. 30,000\n");
    printf("3. Headphones  - Rs. 5,000\n");
}

// 2.
int getChoice() {
    int choice;
    printf("Select a product (1-3): ");
    scanf("%d", &choice);
    return choice;
}

// 3.
float calculateItemTotal(float price, int qty) {
    return price * qty;
}

// 4.
void displayReceipt(float subtotal, float discount, float finalBill) {
    printf("\n====== RECEIPT ======\n");
    printf("Subtotal:      Rs. %.2f\n", subtotal);
```

Untitled1.cpp

```c

// 4.
void displayReceipt(float subtotal, float discount, float finalBill) {
    printf("\n====== RECEIPT ======\n");
    printf("Subtotal:      Rs. %.2f\n", subtotal);
    printf("Discount:      Rs. %.2f\n", discount);
    printf("Total Payable: Rs. %.2f\n", finalBill);
    printf("====================\n");
}

int main() {
    int choice, quantity=0;
    float subtotal = 0, price = 0, discount = 0;
    char cont;

    do {
        showMenu();
        choice = getChoice();

        // Assign price
        if (choice == 1){
            price = 50000;}
        else if (choice == 2) {
        price = 30000;}
        else if (choice == 3) {
        price = 5000;}
        else {
```

Jntitled1.cpp

```c
        price = 5000;}
        else {
            printf("Invalid item choice!\n");
            continue;
        }

        printf("Enter quantity: ");
        scanf("%d", &quantity);

        // Add to subtotal
        subtotal += calculateItemTotal(price, quantity);

        printf("Continue shopping? (y/n): ");
        scanf(" %c", &cont);
    } while (cont == 'y' || cont == 'Y');
    quantity++;

    // Apply 10% discount
    if (subtotal >= 20000) {
        discount = subtotal * 0.10;
    }

    float finalBill = subtotal - discount;

    // Print the bill
    displayReceipt(subtotal, discount, finalBill);
```

Untitled1.cpp

```
71
72      // Print the bill
73      displayReceipt(subtotal, discount, finalBill);
74
75
76      if (choice == 1){
77         printf("you selected laptop\n");
78         printf(" price = 50000");}
79         else if (choice == 2) {
80             printf("you selected smartphone\n");
81         printf(" price = 30000");}
82         else if (choice == 3) {
83             printf("you selected headphones\n");
84         printf(" price = 5000");}
85         else {
86             printf("discount is %d\n",discount);
87
88  }
89  printf("\nquantity is %d\n",quantity);
90  printf("\nsubtotal is %.2f\n",subtotal);
91          printf("discount is %.2f\n",discount);
92          printf("final bill is %.2f\n",finalBill);
93          return 0;
94  }
95
96
97
98
```

```
C:\Users\humai\Desktop\PF T
2. Smartphone  - Rs. 30,000
3. Headphones  - Rs. 5,000
Select a product (1-3): 1
Enter quantity: 1
Continue shopping? (y/n): y

=== SHOPPING MENU ===
1. Laptop      - Rs. 50,000
2. Smartphone  - Rs. 30,000
3. Headphones  - Rs. 5,000
Select a product (1-3): 1
Enter quantity: 1
Continue shopping? (y/n): n

====== RECEIPT ======
Subtotal:      Rs. 100000.00
Discount:      Rs. 10000.00
Total Payable: Rs. 90000.00
====================
you selected laptop
 price = 50000
quantity is 2

subtotal is 100000.00
discount is 10000.00
final bill is 90000.00

--------------------------------
Process exited after 12.57 seconds with return value 0
Press any key to continue . . .
```